Shreejit Verma Final Exam Submission

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MSCF Probability – Final Exam – August 23, 2021

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Instructions:

- You should complete all questions on this exam by writing out the solutions on paper that you will then submit. Do not submit the exam paper itself, even if you are taking the exam in person.
- You will have three hours to complete this exam. No matter when you start the exam, you have to stop at 12:00 PM.
- Notes (except described below), books and calculators/laptops/phones are not allowed.
- A table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k},\ n^k,\ \exp(k),\ n!,\ \Gamma(x),\ \Phi(x),$$

  where $\Phi(x)$ is the CDF of the standard Normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.
- You can write your answers in terms of standard, named CDFs, provided the distribution is **completely specified**. For example, you could write an answer as

$$F(a) - F(b), \text{ where } F \text{ is the CDF of the Normal}(10, 2) \text{ distribution}$$

- You may use 2 sides of a 8.5" by 11" page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.
  Note: The total number of points is 100. You receive 22 points for putting your name on your paper.
- You **must show all of your work**, unless otherwise stated. A correct answer without explanation will receive only minimal or no partial credit. Note that the "easier" a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

PLEASE CHECK THAT YOU HAVE PAGES 1 – 9, THE DISTRIBUTION TABLE, AND THE MATH FACTS

Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

Part I: For each of the following, simply provide the answer. It is not required that any derivation or justification be provided. Each of these is worth three points.

1. Suppose that A and B are disjoint events. Which of the following is always true?

   Choose One:

   (a) $P(A|B) = P(A)$
   (b) A and B are independent
   [✓ (c) $P(A|B) = 0$]
   (d) $P(A|B) \geq P(B)$

$$P(A/B) = \frac{P(A\cap B)}{P(B)} = \frac{0}{P(B)} = 0 \quad \{\because A\,\&\,B \text{ are disjoint}\}$$

2. Suppose that there are n stocks in a particular sector. Let X equal the number of these stocks whose price will go up on the next trading day. If I assume that X has the binomial(n,p) distribution, where $0 < p < 1$, then I am assuming that ...

   Choose One:

   (a) ... the probability of at least one stock going up is p. ×
   [✓ (b) ... the stocks are independent in the sense that given that one stock in the sector has gone up, the probability of another stock in that sector going up is still p.]
   (c) ... exactly np of the stocks will go up.
   (d) ... $P(X = np) \geq P(X = k)$ for any k. ×

3. Suppose that the length of time between "jumps" is well-modelled by the Exponential(λ) distribution. It has been exactly 10.4 hours since the last jump. Which of the following is true? Assume that λ is in units of hour^{-1}.

   Choose One:

   [✓ (a) The remaining time until the next jump has the Exponential(λ) distribution.]
   (b) The expected time until the next jump is $1/\lambda - 10.4$ hours.
   (c) The total time that will elapse from the previous jump until the next jump will have the Pareto($\alpha = 1, \lambda$) distribution.
   (d) The distribution of the remaining time until the next jump depends on how long it has been since the last jump.

4. Suppose that the joint pdf of (X,Y), denoted f_{XY}, is such that $f_{XY}(x,y) > 0$ if and only if $0 < x < 1$ and $0 < y < x$. Which of the following must be true?

   Choose One:

   (a) The marginal density for X, denoted f_X, is such that $f_X(x) > 0$ if and only if $0 < x < 1$.
   [✓ (b) Suppose that $0 < y < 1$. The conditional density for X given Y, denoted $f_{X|Y}$, is such that $f_{X|Y}(x|y) > 0$ if and only if $0 < x < 1$.]
   (c) The random variables X and Y are independent.
   (d) Without further information, it is impossible to claim that any of (a), (b) or (c) are always true.

5. Suppose that (X,Y) is bivariate normal, and that $E(X) = E(Y) = V(X) = V(Y) = 2$. What can be said about the random variable $U = 5X + 3Y$?

   Choose One:

   (a) U definitely has the normal distribution, but nothing else can be said about the mean and variance of U, because $\text{Cov}(X,Y)$ is not given.
   (b) U definitely has the normal distribution, and we know that $E(U) = 16$ and $V(U) = 68$.
   [✓ (c) U definitely has the normal distribution, and we know that $E(U) = 16$ but the variance of U is unknown, because $\text{Cov}(X,Y)$ is not given.]
   (d) We know that $E(U) = 16$, but nothing else can be said about the distribution of U without further information.

$$E(U) = E(5X+3Y) = 5E(X) + 3E(Y) = (5\times 2) + (3\times 2) = 16$$

$$V(U) = V(5X+3Y) = 25V(X) + 9V(Y) + 2\cdot 5\times 3\,\underbrace{\text{Cov}(X,Y)}_{\text{Not Known}}$$

{∵ All linear combinations of bivariate Normal are also Normally distributed (Property 3)}

Part II: Show your work for these questions, partial credit will be given when appropriate. **Be clear in your steps, and justify them.**

1. Suppose that X has the Exponential(λ) distribution. Derive the pdf for $Y = X^2$. **(4)**

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

$$F_Y(y) = P(Y \leq y) = P(x^2 \leq y) = P(\underset{\checkmark}{0 < x < \sqrt{y}} \text{ and } \underset{\times}{-\sqrt{y} \leq x < 0})$$

$$F_Y(y) = 1 - e^{-\lambda\sqrt{y}}$$

$$x^2 < y$$

$$x^2 - y < 0$$

$-\sqrt{y}$ $\sqrt{y}$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = -\frac{1}{\lambda} e^{-\lambda\sqrt{y}} \times \frac{1}{2\sqrt{y}}$$

2. A casino has two slot machines. One pays out 10% of the time, while the other (the "generous machine") pays out 20% of the time. You know that the two machines differ in this way, but you are not sure which is the generous machine. In fact, prior to getting any additional information, you believe that they are each equally likely to be the generous machine.

   Your strategy is to choose one machine at random, and put a coin in to play it. You win on that first play. What is the probability that the machine you just played in the more generous machine? **(5)**

$P(M_1) = 1/2$ {Probability of getting Machine 1}

$P(M_2) = 1/2$ {Probability of getting Machine 2}

$P(W/M_1) = 10\% = 1/10$

$P(W/M_2) = 20\% = \frac{20}{100} = \frac{1}{5}$

We want to calculate

$P(M_2/W)$ = (i.e. Probability that the machine we just played was generous given we won in the first chance)

$$P(M_2/W) = \frac{P(W/M_2)P(M_2)}{P(W/M_1)P(M_1) + P(W/M_2)P(M_2)} = \frac{\frac{1}{5}\times\frac{1}{2}}{\frac{1}{10}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{5}} = \frac{\frac{1}{10}}{\frac{3}{20}} = \frac{2}{3} \text{ Ans}$$

3. Suppose that I am going to toss a six-sided die twice. (You can assume the die is fair, but it actually does not matter for this question.) A natural way to define the sammple space in this case is using 36 outcomes as follows:

$$\Omega = \{\{1,1\},\{1,2\},\ldots,\{6,5\},\{6,6\}\}$$

   Define the random variable Y to equal the sum of the two tosses.

   Write out **one** subset of Ω that is **not** in $\sigma(Y)$, and then explain why that set is not in $\sigma(Y)$. **(4)**

$Y = t_1 + t_2$

t_1 t_2

Y=2 (1,1) (1,2) (1,3) (1,4) (1,5) (1,6)

Y=3 (2,1) (2,2) (2,3) (2,4) (2,5) (2,6)

Y=4 (3,1) (3,2) (3,3) (3,4) (3,5) (3,6)

Y=5 (4,1) (4,2) (4,3) (4,4) (4,5) (4,6)

Y=6 (5,1) (5,2) (5,3) (5,4) (5,5) (5,6)

Y=7 (6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

Y=8 Y=9 Y=10 Y=11 Y=12

Y
- 2 {(1,1)} P(Y=2) = 1/36
- 3 {(2,1),(1,2)} P(Y=3) = 2/36
- 4 {(3,1),(2,2),(1,3)} P(Y=4) = 3/36
- 5 {(4,1),(3,2),(2,3),(1,4)} P(Y=5) = 4/36
- 6 {(5,1),(4,2),(3,3),(2,4),(1,5)} P(Y=6) = 5/36
- 7 {(6,1),(5,2),(4,3),(3,4),(2,5),(1,6)} P(Y=7) = 6/36
- 8 {(6,2),(5,3),(4,4),(3,5),(2,6)} P(Y=8) = 5/36
- 9 {(6,3),(5,4),(4,5),(3,6)} P(Y=9) = 4/36
- 10 {(6,4),(5,5),(4,6)} P(Y=10) = 3/36
- 11 {(6,5),(5,6)} P(Y=11) = 2/36
- 12 {(6,6)} P(Y=12) = 1/36

We don't have specific Probability of getting a case where Y has multiple possibilities For example P(4,3) is not known we only know P(Y=7) but not P(4,3).

4. Suppose that (X,Y) has joint pdf

$$f_{(X,Y)}(x,y) = \begin{cases} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)}\, x^{\alpha-1}\, y^{\beta-1}, & \text{if } x > 0,\ y > 0, \text{ and } x + y < 1 \\ 0, & \text{otherwise} \end{cases}$$

   where $\alpha > 0$ and $\beta > 0$.
   Note: Remember that $\Gamma(z+1) = z\Gamma(z)$ for any $z > 0$.

   (a) Show that the (marginal) distribution of X is Beta$(\alpha, \beta+1)$. **(4)**

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dy = \int_0^{1-y} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} y^{\beta-1}\,dy = \left[\frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} x^{(\alpha-1)} \frac{y^\beta}{\beta}\right]_0^{1-y}$$

$$= \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)} (1-y)^{\alpha-1} y^{\beta} = \text{Beta}(\alpha, \beta+1)$$

llyr

$$= \text{Beta}(\alpha+1, \beta)$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dx = \int_0^{1-x} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} y^{\beta-1}\,dx = \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} \frac{x^\alpha}{\alpha}(1-x)^{\beta-1} = \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha+1)\Gamma(\beta)} x^{\alpha}(1-x)^{\beta-1} = \text{Beta}(\alpha+1, \beta)$$

(b) What is $E(Y)$? **(3)**

$$E(Y) = \int_{-\infty}^{\infty}\boxed{\int_{-\infty}^{\infty} y\, f(x,y)\,dx}\,dy$$

We know, $f_Y(y) = \text{Beta}(\alpha+1, \beta)$

so $E(Y) = \frac{\alpha+1}{\alpha+\beta+1}$

$$E(X) = \frac{\alpha}{\alpha+\beta+1}$$

$$E(Y) = \frac{\alpha+1}{\alpha+\beta+1}$$

(c) What is the covariance between X and Y? **(3)**

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$= \underline{\qquad} - \frac{\alpha}{(\alpha+\beta+1)} \times \frac{\alpha+1}{(\alpha+\beta+1)}$$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f_{X,Y}(x,y)\,dx\,dy = \int_0^{1-x}\int_0^{1-y} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha} y^{\beta}\,dx\,dy = \int_0^{1-x} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)} \frac{(1-y)^{\alpha}\, y^{\beta+1}}{(\beta+1)}\,dx$$

(d) Derive the conditional distribution of Y given X. In other words, what is $f_{Y|X}$? **(4)**

$$f_{Y/X}(y/x) = f_{X,Y}(x,y) / f_Y(y)$$

5. Suppose that there are four envelopes – two are green and two are red. There are also four cards – two are green and two are red. The cards are randomly assigned to the envelopes in such a way that each card is equally likely to end up in each of the four envelopes, but each envelope has exactly one card in it. What is the probability that all four envelopes end up with cards of matching colors? **(4)**

4 Envelopes (E_G, E_G, E_R, E_R) 12 34

4 Cards (C_G, C_G, C_R, C_R)

$\overline{E_G}$ $\overline{E_G}$ $\overline{E_R}$ $\overline{E_R}$

Desired Event $= \frac{{}^2C_1 \times {}^1C_1 \times {}^2C_1 \times {}^1C_1}{2!} = 2$

Total Events $= \frac{4!}{2!2!} = 6$

$$\frac{4}{24} = \frac{1}{6} \text{ Ans}$$

4! = 24

14 32
14 23
1 2 3 4
1 2 4 3
1 3 4 2
1 3 2 4
2 1 3 4
2 1 4 3
2 3 4 1
2 3 1 4
2 4 1 3
2 4 3 1

3 1 2 4
3 1 4 2
3 2 4 1
3 2 1 4
3 4 1 2
3 4 2 1
4 1 2 3
4 1 3 2
4 2 3 1
4 2 1 3
4 3 1 2
4 3 2 1

6. There are n "experts" that each day predict whether or not a particular stock will go up or down. In reality, each of them is flipping a fair coin to make their "prediction." There is a fixed period of m trading days during which their performance will be monitored. What is the probability that there is at least one of the experts that predicts correctly every day of the m trading days? **(5)**

This is a type of Binomial distribution where $p = \frac{1}{2}$, n is a sequence of independent Bernoulli trials & m is total observed events

$$P\left(\text{at least one expert is correct for every day of the m trading days}\right) = 1 - P\left(\text{no expert is correct for m trading days}\right) = 1 - \left(1 - \tfrac{1}{2}\right)^m$$

7. Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d., each with the Exponential(λ) distribution. You can assume that $n > 50$.

   (a) Show that $X_{(1)}$, which is the minimum of the X_i has the Exponential(β) distribution. What is the parameter β as a function of λ? **(4)**

$$P(X_{(1)} \leq x_i) = 1 - P(X_{(1)} > x_i) = 1 - P(X_1 > X_2 \text{ and } X_1 > X_3 \ldots X_1 > X_n) = 1 - (e^{-\lambda t})^n = 1 - e^{-n\lambda t}$$

$X_{(1)} \sim$ exponential $(n\lambda)$

$\beta = n\lambda$

   (b) Approximate the probability that $\overline{X}$ is larger than c. As usual, $\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i$. **(4)**

$$P(\overline{X} > c) = 1 - F_X(c) = 1 - (1 - e^{-n\lambda c}) = e^{-n\lambda c}$$

8. Suppose that $X_1, X_2, \ldots X_n$ are i.i.d. random variables, each with the Geometric(p) distribution. Let $Y = \sum_{i=1}^n X_i$.

   (a) The table on the last page of the exam includes the *negative binomial*(r,p) *distribution*. Show that Y has the negative binomial distribution with $r = n$. **(4)**

$$Y = \sum_{i=1}^n x_i = \sum_{i=1}^n (1-p)^{x_i} p$$

$$M_X(t) = \sum_{K=1}^{n} \left((1-p)^{x_i} p\right)^{x \to RV} \frac{e^{-\lambda} \lambda^K}{K!} \quad \text{Rule of Lazy Stat.}$$

$$=$$

Not able to figure out how it converts to this $\rightarrow \binom{n+x-1}{x} p^n (1-p)^x$

$$\frac{(n+x-1)!}{x!\,(n-1)!}$$

   (b) Which of the following is an appropriate practical interpretation of Y? **(3)**

   Choose One:

   i. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of failures before seeing the n^{th} success.
   ii. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of trials before seeing the n^{th} success.
   iii. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of failures in the first n trials.
   iv. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of successes in the first n trials.

9. Suppose that $X_1, X_2, \ldots, X_n$ are the times between successive events in a Poisson process with rate λ. Thus, $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ). A good estimator for the probability that there will be zero events during a period of length one time unit is $g(\overline{X}) = \exp(-1/\overline{X})$, where, as usual, $\overline{X}$ is the sample mean of the X_i.

   Use the Delta method to approximate the distribution of $g(\overline{X})$. You can assume that n is larger than 50. **(4)**

   Note: Your expressions for the mean and variance of the normal distribution should only be in terms of λ and n.

$g(\bar{x}) = \exp\left(-\frac{1}{\bar{x}}\right)$ {$\bar{x}$ is mean of x_i}

0 occurrence means $e^{-\lambda}$

$\Rightarrow e^{-\lambda} = e^{-1/\bar{x}} \Rightarrow \frac{-1}{\bar{x}} = -\lambda$ {by comparision}

$\Rightarrow \bar{x} = \frac{1}{\lambda}$ or $\lambda = \frac{1}{\bar{x}}$

$$g(\bar{x}) = e^{(-1/\bar{x})} = e^{(-1/x)} = e^{-\lambda}$$

$$g'(x) = \frac{e^{-1/x}}{x^2} = -\lambda^2 e^{-\lambda}$$

and variance $(g'(\mu))^2 \frac{\sigma^2}{n} = \lambda^4 e^{-2\lambda} \times \frac{\lambda^2}{n}$

So by Δ method $\frac{1}{\bar{x}} \simeq N\left(e^{-\lambda}, \frac{\lambda^6 e^{-2\lambda}}{n}\right)$

10. We saw in lecture that for a Poisson process with rate λ, the number of events during a time interval of length t has the Poisson(λt) distribution. (Of course, this statement is based on t and $1/\lambda$ being in the same units.)

   (a) Suppose that calls arrive as a Poisson process with rate $\lambda = 10.5$ per hour. What is the probability exactly one call will arrive during a chosen ten minute period? **(4)**

The probability of r occurrence in a given time interval of length t is $\frac{\lambda^r t}{r!} e^{-\lambda t}$

{∵ λ = 10.5 per hour, t = 10 minute = $\frac{1}{6}$ hour.}

$$\text{Probability of 1 occurrence} = \frac{\lambda^1 t}{1!} \times e^{-\lambda/6} = \left(\frac{10.5}{6}\right)^1 \times e^{-\frac{10.5}{6}} = \frac{10.5}{6} \times e^{-\frac{10.5}{6}}$$

   (b) Imagine that one is not comfortable with assuming that the Poisson process is a reasonable model for calls arriving into the call center. Instead, we want to imagine that the rate of calls varies from hour to hour in an unpredictable manner.

   In particular, we will model the arrival of calls as follows. For each hour period, i.e., from 1:00 to 2:00, from 2:00 to 3:00, and so forth, the rate at which calls arrive follows the Gamma(α, β) distribution. The rate is still in units of hour^{-1}. The rate is independent from hour to the next, and the number of calls is independent from one hour to the next. To be clear: Although it is randomly chosen, the rate is constant during the one-hour period.

   What is the expected number of calls during the hour 2:00 to 3:00 PM? **(4)**